

Isaac S. Garee

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EDUCATION

University of Cincinnati

Bachelor of Science, Industrial and Systems Engineering

Minors, Mathematics and Statistics

Expected May 2027

GPA: 3.5

- Primary Coursework: Linear Algebra, Deterministic Systems Models, Probabilistic Systems Models, Data Science, Python Programming, Probability and Statistics

EXPERIENCE

University of Cincinnati | Cincinnati, Ohio

Apr 2025 – Present

Undergraduate Researcher, Advised by Prof. Tommaso Giovannelli

- Engineered a novel mixed-integer bilevel optimization Neural Architecture Search (NAS) framework, delivering a 47% speedup over the current baselines while maintaining competitive accuracy
- Designed and implemented a PyTorch codebase from scratch including finite-difference gradients, Hessian-free updates, and line-search functions, enabling scalable experimentation across architecture search spaces
- Benchmarked algorithm convergence and stability against existing NAS baselines, evaluating architecture quality, training dynamics, and validation accuracy
- Proved key theoretical properties of the mixed-integer bilevel algorithm in collaboration with faculty advisor, contributing to a research manuscript currently in preparation

Kaivac | Hamilton, Ohio

Jan 2024 – Dec 2024

Industrial and Systems Engineering Co-op

- Reduced inbound inspection process time by 20% by identifying bottlenecks through value stream mapping and implementing targeted process improvements
- Delivered Power BI dashboards that unified cross-functional production data, enabling faster management decisions and eliminating manual reporting workflows
- Deployed control charts and SPC metrics across manufacturing lines, introducing real-time quality monitoring where none previously existed
- Expanded data tracking infrastructure to capture previously unmeasured process variables, generating new insights that sustained continuous improvement efforts

PROJECTS

Risk Assessment Model for Osteoradionecrosis (ORN) | Python / Machine Learning

Apr 2024 – May 2024

- Developed a population-based risk assessment model for ORN risk prediction model from a dataset of 1,259 head and neck cancer patients, applying K-means clustering to dose-volume histograms and pre-radiation dental extraction status to generate distinct patient risk categories
- Validated model accuracy across all risk categories using 95% confidence intervals, confirming that actual ORN outcomes fell within predicted ranges on a held-out test set

SKILLS

Programming and Tools: Python, R, PyTorch, NumPy, SciPy, Pandas, LaTeX, Gurobi, Excel, Jupyter Notebooks

Technical Skills: Bilevel Optimization, Machine Learning, Value Stream Mapping, Stochastic Modeling

ACTIVITIES & HONORS

Industrial & Systems Engineering Student Association | Secretary

Jan 2024 – Jan 2025

- Coordinated meeting logistics, recorded minutes, and organized guest speaker sessions for a 50+ member engineering organization

Shidecker Memorial Golden Eagle Scholarship

Aug 2022 – May 2023

- Awarded a scholarship from Shidecker Memorial Foundation in recognition of commitment to both academic excellence and community involvement